Jan 2022 Q2 Sample Solutions.

Q2 a) Differentiate using first principles — must show all working using first principle

$$\frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \quad \text{(Formula)}$$

$$f(x) = y = 2x^2 - 3x + 6$$

$$\begin{aligned} f(x+h) &= 2(x+h)^2 - 3(x+h) + 6 \\ &= 2(x+h)(x+h) - 3x - 3h + 6 \\ &= 2(x^2 + xh + xh + h^2) - 3x - 3h + 6 \\ &= 2(x^2 + 2xh + h^2) - 3x - 3h + 6 \\ &= 2x^2 + 4xh + 2h^2 - 3x - 3h + 6 \end{aligned}$$

$$\frac{dy}{dx} = \lim_{h \to 0} \frac{(2x^2 + 4xh + 2h^2 - 3x - 3h + 6) - (2x^2 - 3x + 6)}{h}$$

$$= \lim_{h \to 0} \frac{2x^2 + 4xh + 2h^2 - 3x - 3h + 6 - 2x^2 + 3x - 6}{h}$$

$$= \lim_{h \to 0} \frac{4xh + 2h^2 - 3h}{h}$$

$$= \lim_{h \to 0} 4x + 2h - 3 \qquad (\text{sub in } h = 0)$$

$$= 4x + 2(0) - 3$$

$$dy/dx = \boxed{4x - 3}$$

b) i) $y = \frac{5x}{x+2}$ (u over v)

use Quotient Rule:

$$\frac{df}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

$u = 5x \quad \frac{du}{dx} = 5$

$v = x+2 \quad \frac{dv}{dx} = 1$

sub into the formula

$$\frac{df}{dx} = \frac{(x+2)(5) - (5x)(1)}{(x+2)^2}$$

$$= \frac{5x+10-5x}{(x+2)^2}$$

$$\frac{df}{dx} = \boxed{\frac{10}{(x+2)^2}}$$

ii) $y = (3x+1)(4x-5)$ (u, v)

use Product Rule:

$$\frac{df}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$$

$u = (3x+1) \quad du/dx = 3$

$v = (4x-5) \quad dv/dx = 4$

sub into the formula

$$\frac{df}{dx} = (3x+1)(4) + (4x-5)(3)$$

$$= 12x+4+12x-15$$

$$\frac{df}{dx} = \boxed{24x-11}$$

c) $y = x^3 - 12x + 2$ use Power Rule: $\frac{dy}{dx} = nx^{n-1}$

i) $\frac{dy}{dx} = 3x^2 - 12$

$\frac{d^2y}{dx^2} = 6x$

ii) if there is a turning point then $\frac{dy}{dx} = 0$

let $\frac{dy}{dx} = 0$ and solve for x

$\therefore \quad 3x^2 - 12 = 0$

$3x^2 = 12$

$x^2 = \frac{12}{3}$

$x^2 = 4$

$x = 2$ or $x = -2$

To find which is the Max / Min point:

$\frac{d^2y}{dx^2} = 6x$ when $x = 2$: $6(2) = 12$ (positive) $\therefore$ Min point

when $x = -2$: $6(-2) = -12$ (negative) $\therefore$ Max point

coordinates : when $x = 2$ $y = 2^3 - 12(2) + 2$

$= -14$

$\Rightarrow$ Min turning point = $\boxed{(2, -14)}$

when $x = -2$ $y = -2^3 - 12(-2) + 2$

$= 18$

$\Rightarrow$ Max turning point = $\boxed{(-2, 18)}$

d) area enclosed by the curve (definite integral)

$y = x + 3x^2$ $\quad x = 0$ or $x = 2$

$\therefore \int_0^2 (x + 3x^2) \cdot dx$ (definite integral)

$= \left[\frac{x^2}{2} + \frac{3x^3}{3} \right]_0^2 = \left[\frac{x^2}{2} + x^3 \right]_0^2$

$= \left(\frac{2^2}{2} + 2^3 \right) - \left(\frac{0^2}{2} + 0^3 \right)$

$= 10 - 0$

$= \boxed{10 \text{ square units}}$